

## Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to real-world scenarios.
4. To foster independent learning and research skills.
5. To encourage active participation and engagement in class discussions and activities.
6. To prepare students for further academic pursuits or professional careers in the field.
7. To promote the development of effective communication and collaboration skills.
8. To instill a sense of ethical responsibility and social awareness.

### **1. To enable the students to understand about the Chemistry of Atomic and Molecular structure, Chemistry of:**

- Atomic structure refers to the fundamental structure of an atom, including its nucleus and electrons.
- The nucleus is made up of protons and neutrons, and is surrounded by electrons in shells or orbitals.
- The number of protons in the nucleus determines the atomic number and the element to which the atom belongs.
- The arrangement of electrons in the atom's outermost shell determines its chemical properties and reactivity.
- Molecular structure refers to the arrangement of atoms in a molecule and the chemical bonds that hold them together.
- The type and strength of chemical bonds, as well as the shape of the molecule, determine its properties and reactivity.
- Understanding the chemistry of atomic and molecular structure is essential for predicting and explaining chemical reactions and the properties of substances.

## Advanced Materials

Advanced materials refer to materials that have been developed through the incorporation of new or improved properties, making them superior to traditional materials in terms of performance or sustainability. Some examples of advanced materials include liquid crystals, nanomaterials, graphite and fullerenes, and green chemistry.

### Liquid Crystals

- Liquid crystals are a state of matter that has properties between those of conventional liquids and those of solid crystals.

- They can flow like a liquid, but have the long-range order of a solid crystal.
- Liquid crystals are used in a variety of applications, including displays, sensors, and thermometers.

## **Nanomaterials**

- Nanomaterials are materials with at least one dimension in the nanometer range (1-100 nm).
- They exhibit unique properties due to their small size and high surface area to volume ratio.
- Nanomaterials have a wide range of applications, including electronics, medicine, and energy.

## **Graphite and Fullerenes**

- Graphite is a form of carbon that consists of layers of carbon atoms arranged in a hexagonal lattice.
- Fullerenes are a family of carbon allotropes that are composed entirely of carbon, in the form of a hollow sphere, ellipsoid, or tube.
- Graphite and fullerenes have a variety of applications, including lubricants, batteries, and catalysts.

## **Green Chemistry**

- Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances.
- It aims to minimize the environmental impact of chemical processes and products.
- Green chemistry has a wide range of applications, including the development of safer and more sustainable products and processes.

## **2. To enable the students to understand and apply the detailed concepts of spectroscopic techniques:**

- Spectroscopy is the study of the interaction between matter and electromagnetic radiation.
- Spectroscopic techniques are used to analyze the chemical composition and structure of materials.
- There are several types of spectroscopy, including infrared, ultraviolet-visible, and nuclear magnetic resonance spectroscopy.
- Infrared spectroscopy measures the absorption of infrared radiation by molecules, which can provide information about the functional groups present in a molecule.
- Ultraviolet-visible spectroscopy measures the absorption of ultraviolet and visible light by molecules, which can provide information about the electronic structure of a molecule.

- Nuclear magnetic resonance spectroscopy measures the absorption of radiofrequency radiation by atomic nuclei in a magnetic field, which can provide detailed information about the structure and dynamics of molecules.
- Spectroscopic techniques can be applied in a wide range of fields, including chemistry, biology, physics, and materials science.
- To effectively use spectroscopic techniques, students must have a strong understanding of the underlying principles and be able to interpret the data obtained from these techniques.

## Stereochemistry

Stereochemistry is the branch of chemistry that deals with the spatial arrangement of atoms in molecules and the effects of this arrangement on the chemical and physical properties of the compounds.

- Stereoisomers are isomers that have the same molecular formula and sequence of bonded atoms, but differ in the three-dimensional orientations of their atoms in space.
- Stereoisomers can be classified into two main types: enantiomers and diastereomers.
- Enantiomers are non-superimposable mirror images of each other. They have identical physical and chemical properties, except for their ability to rotate plane-polarized light in opposite directions.
- Diastereomers are stereoisomers that are not mirror images of each other. They have different physical and chemical properties.
- The presence of chiral centers, or asymmetric carbon atoms, is a common cause of chirality in organic molecules. A chiral center is a carbon atom that is bonded to four different groups.
- The Cahn-Ingold-Prelog (CIP) priority rules are used to assign the absolute configuration of chiral centers as either R or S.
- The relative configuration of diastereomers can be determined using methods such as X-ray crystallography, NMR spectroscopy, and chemical correlation.

Stereochemistry plays a crucial role in many areas of chemistry, including organic synthesis, biochemistry, and medicinal chemistry. The correct stereochemical configuration of a molecule can have a significant impact on its biological activity and properties. Understanding stereochemistry is essential for identifying and characterizing compounds and for designing and synthesizing new molecules with desired properties.

### **3. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion**

Electrochemistry is the branch of chemistry that deals with the chemical changes produced by electricity and the production of electricity by chemical changes.

Some of the key concepts in electrochemistry include:

1. **Redox reactions:** These are reactions in which electrons are transferred from one species to another. The species that loses electrons is oxidized, while the species that gains electrons is reduced.
2. **Electrochemical cells:** These are devices that convert chemical energy into electrical energy or vice versa. There are two types of electrochemical cells: galvanic cells and electrolytic cells.
3. **Batteries:** A battery is a device that stores chemical energy and converts it into electrical energy. Batteries consist of one or more electrochemical cells connected in series or parallel.
4. **Corrosion:** Corrosion is the process by which a metal is gradually destroyed by a chemical reaction with its environment. This process is an electrochemical reaction that involves the transfer of electrons.

By understanding these concepts, students will be able to apply them to real-world situations, such as designing and using batteries, preventing corrosion, and carrying out electrochemical reactions. It is important for students to have a strong foundation in these concepts in order to succeed in their studies and future careers.

## Chemistry of Engineering Materials: Cement

Cement is a binding material that is used in construction to set, harden, and adhere to other materials. The chemical composition of cement depends on the raw materials used in its manufacturing. These raw materials include lime, silica, alumina, and iron oxide.

The cement clinker is produced by igniting a mixture of raw materials, one of which is composed mainly of calcium carbonate and the other of aluminosilicates and oxides of aluminum and iron. The following complex mineralogical compounds constitute the cement clinker.

The raw materials for cement production are limestone (calcium), sand or clay (silicon), bauxite (aluminum), and iron ore. These may include shells, chalk, marl, shale, clay, blast furnace slag, and slate. Chemical analysis of cement raw materials provides insight into the chemical properties of cement.

Cements may be used alone, but the normal use is in mortar and concrete in which the cement is mixed with an inert material known as aggregate. Mortar is cement mixed with sand or crushed stone that must be less than approximately 5 mm in size.

Researchers have developed methods to monitor the cement hydration process in real-world conditions. The concrete world that surrounds us owes its shape

and durability to chemical reactions that start when ordinary Portland cement is mixed with water.

**4. To enable the students to understand and apply detailed concepts of water source, water impurities, hardness**

- Water is a vital resource for life and is obtained from various sources such as rivers, lakes, wells, and oceans.
- Water from these sources may contain impurities such as dissolved minerals, organic matter, and microorganisms.
- Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It is usually expressed in terms of the equivalent quantity of calcium carbonate.
- Water hardness can be classified as temporary or permanent. Temporary hardness is caused by the presence of bicarbonate ions and can be removed by boiling. Permanent hardness is caused by the presence of sulfate, chloride, and nitrate ions and cannot be removed by boiling.
- Water impurities and hardness can have various effects on human health, industrial processes, and household appliances. It is important to monitor and treat water to ensure its safety and suitability for use.

## **Water and Boiler Troubles Used in Industry**

- Many commercial, manufacturing, and industrial processes use water as a primary medium. These include the use of boilers and steam raising plant, cooling towers, cooling systems, and other closed circuit systems.
- Some of the most common industrial and commercial boiler problems include missing insulation, irregular or infrequent maintenance, water leaks, pressure-related issues, and scale buildup and blockages.
- Many boiler problems start with the water quality. Common boiler feed water issues or problems with the condensate return water can lead to severe damage to the unit.

## **Analysis of Coal & Determination of Calorific Values**

- Calorific value is a measure of the amount of energy produced from a unit weight of coal when it is combusted in oxygen.
- A measured sample of coal is completely combusted in a bomb calorimeter, which is a device for measuring heat (ASTM method D5865-12; American Society for Testing and Materials, 2013, p. 648–666).
- Research has been conducted on the applicability of complex thermal analysis techniques (TG/DTG, DTA) in the determination of calorific values of coals of different origin. The calorific values obtained by thermal anal-

ysis are compared with an adiabatic bomb calorimeter by the standard ASTM method.

#### 5. To enable the students to understand detailed concepts related to polymers, Polymerization, Polymer Blends

- **Polymers** are large molecules composed of repeating structural units, typically connected by covalent chemical bonds. They can be natural or synthetic and have a wide range of properties and applications.
- **Polymerization** is the process of chemically bonding monomers (small molecules) together to form a polymer. There are two main types of polymerization: addition polymerization and condensation polymerization.
- **Addition polymerization** involves the joining of monomers with unsaturated bonds, such as ethylene, to form a polymer. The process is initiated by a free radical, an anion, or a cation, which adds to the unsaturated monomer to form a new reactive site.
- **Condensation polymerization** involves the elimination of a small molecule, such as water, during the polymerization process. This type of polymerization typically occurs between monomers with two functional groups, such as an alcohol and a carboxylic acid, to form a polymer with an ester linkage.
- **Polymer blends** are mixtures of two or more polymers that are physically mixed but not chemically bonded. Blending polymers can result in materials with improved properties, such as increased strength, flexibility, or heat resistance. The properties of the blend depend on the properties of the individual polymers, their ratio, and the degree of mixing.

## Polymer Composites

Polymer composites are materials made of a polymer matrix reinforced with fibers, particles, or other fillers. The polymer matrix is usually a thermoset or thermoplastic resin, while the reinforcement can be made of glass, carbon, or natural fibers, among others.

Some of the advantages of using polymer composites include: - High strength-to-weight ratio: Polymer composites can be designed to have high strength and stiffness while being lightweight. - Corrosion resistance: Polymer composites are resistant to many chemicals and environmental conditions, making them suitable for use in harsh environments. - Design flexibility: Polymer composites can be molded into complex shapes, allowing for greater design flexibility. - Durability: Polymer composites can have long service lives due to their resistance to fatigue and wear.

Some of the applications of polymer composites include: - Aerospace: Polymer composites are used in the manufacture of aircraft components due to their high strength-to-weight ratio and resistance to fatigue. - Automotive: Poly-

mer composites are used in the manufacture of car body panels, bumpers, and other components due to their light weight and design flexibility. - Construction: Polymer composites are used in the construction of bridges, buildings, and other structures due to their strength, durability, and resistance to corrosion. - Sports equipment: Polymer composites are used in the manufacture of sports equipment such as tennis rackets, golf clubs, and bicycles due to their light weight and high strength.

## Content Contact

Content contact refers to the interaction between the content of a message and the audience receiving it. This can include the following points:

1. The relevance of the content to the audience: The content should be relevant and interesting to the audience, addressing their needs and interests.
2. The clarity of the content: The content should be clear and easy to understand, avoiding jargon or technical terms that may be unfamiliar to the audience.
3. The tone of the content: The tone of the content should be appropriate for the audience, whether it is formal or informal, serious or lighthearted.
4. The organization of the content: The content should be well-organized and structured, with a clear flow of information and logical progression of ideas.
5. The use of visuals: The use of visuals, such as images or graphs, can help to enhance the content and make it more engaging for the audience.
6. The credibility of the content: The content should be credible and trustworthy, with accurate and reliable information that is supported by evidence.

Overall, effective content contact involves creating a message that is engaging, informative, and relevant to the audience, with the goal of achieving a desired outcome or response.

## Hours

- An hour is a unit of time conventionally defined as 1/24th of a day and scientifically reckoned as 3,599–3,601 seconds, depending on conditions.
- The hour was initially established in the ancient Near East as a variable measure of 1/12 of the night or daytime.
- The seasonal, temporal, or unequal hour was established in the ancient Near East as 1/12 of the night or daytime.
- The hour was subsequently divided into 60 minutes, each of 60 seconds.
- The hour is commonly symbolized using the abbreviation “h” or “hr”.
- The time of day is typically expressed in terms of hours.

- The 24-hour clock is a convention of timekeeping in which the day runs from midnight to midnight and is divided into 24 hours, indicated by the hours passed since midnight, from 0 to 23.
- The 12-hour clock is a time convention in which the 24 hours of the day are divided into two periods: a.m. (from the Latin, ante meridiem, meaning before midday) and p.m. (post meridiem, meaning after midday).
- Each period consists of 12 hours numbered: 12 (acting as zero), 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11.
- The 24-hour clock is used in most countries, particularly those in Europe, while the 12-hour clock is commonly used in countries such as the United States, Canada (except Québec), Australia, New Zealand, India, Pakistan, the Philippines, Colombia, and others.
- The 24-hour clock is also commonly used in military and transportation contexts.
- The International System of Units (SI) defines the second as the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium-133 atom.
- The hour is not an SI unit, but it is accepted for use with SI units.
- The hour is commonly used in various fields, including science, medicine, transportation, and law.

## Unit-1: 8

1. The number 8 is an even integer that comes after 7 and before 9.
2. In the base 10 number system, 8 is the cube of 2.
3. The number 8 is considered lucky in some cultures, such as Chinese culture, where it is associated with wealth and prosperity.
4. In music, an octave is the interval between one musical pitch and another with half or double its frequency, and is named after the Latin word for "eight".
5. In computer science, a byte is typically made up of 8 bits.
6. In the RGB color model, each color channel is often represented by 8 bits, allowing for 256 different shades of each primary color.
7. In sports, the number 8 is often associated with the position of the center in basketball and the scrum-half in rugby.
8. In the periodic table, the element with the atomic number 8 is oxygen.

## Atomic and Molecular Structure: Molecular orbitals of diatomic molecules, Bond Order

Molecular orbitals are formed when atomic orbitals combine. In diatomic molecules, the atomic orbitals of two atoms combine to form molecular orbitals. These molecular orbitals can be bonding, non-bonding, or anti-bonding.

- Bonding molecular orbitals are lower in energy than the atomic orbitals from which they are formed. Electrons in bonding orbitals stabilize the



molecule.

- Non-bonding molecular orbitals have the same energy as the atomic orbitals from which they are formed. Electrons in non-bonding orbitals do not affect the stability of the molecule.
- Anti-bonding molecular orbitals are higher in energy than the atomic orbitals from which they are formed. Electrons in anti-bonding orbitals destabilize the molecule.

The bond order is a measure of the strength of a chemical bond. It is calculated as the number of bonding electrons minus the number of anti-bonding electrons, divided by two. A higher bond order indicates a stronger bond.

For example, in the case of a diatomic molecule such as O<sub>2</sub>, the bond order is 2, indicating a double bond between the two oxygen atoms. In the case of N<sub>2</sub>, the bond order is 3, indicating a triple bond between the two nitrogen atoms.

## Magnetic Characteristics and Numerical Problems

Magnetic characteristics refer to the properties of a material that determine its behavior in the presence of a magnetic field. These characteristics include:

1. **Permeability:** The ability of a material to conduct a magnetic field. It is defined as the ratio of the magnetic flux density in the material to the magnetic field strength.
2. **Magnetic susceptibility:** The degree to which a material can be magnetized in an external magnetic field. It is defined as the ratio of the magnetization of the material to the applied magnetic field strength.
3. **Retentivity:** The ability of a material to retain its magnetization after the external magnetic field is removed.
4. **Coercivity:** The strength of the external magnetic field required to reduce the magnetization of a material to zero.

Numerical problems involving magnetic characteristics typically involve calculating the magnetic field strength, magnetic flux density, magnetization, and other related quantities in various scenarios. These problems may require the use of formulas and equations to solve for the unknown quantities.

Here is an example of a numerical problem involving magnetic characteristics:

**Problem:** A magnetic material has a magnetic susceptibility of 0.8 and is placed in an external magnetic field of strength 500 A/m. Calculate the magnetization of the material.

**Solution:** The magnetization of the material can be calculated using the formula  $M = \chi H$ , where  $M$  is the magnetization,  $\chi$  is the magnetic susceptibility,

and  $H$  is the magnetic field strength.

Substituting the given values into the formula, we get:

$$M = \chi H = (0.8)(500 \text{ A/m}) = 400 \text{ A/m}$$

Therefore, the magnetization of the material is 400 A/m.

### **Chemistry of Advanced Materials:**

1. Advanced materials are materials that exhibit superior properties compared to traditional materials.
2. These properties can include strength, durability, conductivity, and reactivity.
3. The chemistry of advanced materials involves the study of the composition, structure, and properties of these materials.
4. This field of study also includes the development of new materials and the improvement of existing materials.
5. Some examples of advanced materials include nanomaterials, biomaterials, and smart materials.
6. Nanomaterials are materials with structures on the nanoscale, which can exhibit unique properties due to their small size.
7. Biomaterials are materials that are designed to interact with biological systems, such as implants or drug delivery systems.
8. Smart materials are materials that can change their properties in response to external stimuli, such as temperature or light.
9. The development of advanced materials requires a deep understanding of the underlying chemistry, as well as the ability to manipulate materials at the molecular level.
10. The field of advanced materials chemistry is constantly evolving, with new materials and applications being discovered all the time.

### **Liquid Crystals**

Liquid crystals are a state of matter that has properties between those of conventional liquids and those of solid crystals. They are called liquid crystals because they exhibit the mechanical properties of a liquid, but have the optical properties of a single crystal.

#### **Types of Liquid Crystals**

Liquid crystals can be classified into two types: thermotropic and lyotropic. Thermotropic liquid crystals are obtained by melting a crystalline solid, while lyotropic liquid crystals are found in certain colloidal solutions, such as aqueous solutions of tobacco mosaic virus and certain polymers.

#### **Applications of Liquid Crystals**

Liquid crystals have a wide range of commercial applications. They are used in liquid crystal displays (LCDs) in digital watches, calculators, and computer and video displays. Apart from display-related applications, liquid crystals are also attractive for several kinds of sensing applications. Some other applications would be in the areas of lasers and medical diagnostics.

### Industry

The flat panel display is the most recognized device where liquid crystals are widely used. The use of liquid crystals in display technology stems from the nature of chirality and the high electro-optic coefficient of these mediums, thereby making them significantly advantageous. There are also ongoing developments in the industry towards liquid crystal-based smart fabrics.

## Important Materials Used as Liquid Crystals

Liquid crystals are substances that blend the structures and properties of the normally disparate liquid and crystalline solid states. They can be divided into 3 main types: thermotropic, lyotropic, and metallotropic.

- The amphiphilic molecules generally used in the formation of lyotropic liquid crystals can be long-chain alcohols, deoxyribonucleic acid (DNA), gelatin, surfactants, and membrane phospholipids, whereas, the hydrophilic solvent is usually water .
- Many biological materials form liquid crystals. Myelin, a fatty material extracted from nerve cells, was the first intensively studied liquid crystal. The tobacco mosaic virus, with its rodlike shape, forms a nematic phase.
- Liquid crystals can be first organized into thermotropic (temperature dependent) and lyotropic (concentration dependent). Within thermotropic liquid crystals there are high molar mass materials (suitable for polymer) and low molar mass materials.

### Graphite and Fullerene; Introduction, Structure and applications

- **Introduction:** Graphite and fullerene are two forms of carbon that have unique structures and properties. Graphite is a naturally occurring mineral, while fullerene is a synthetic molecule.
- **Structure:** Graphite has a layered structure in which hexagonal rings are connected to each other. The fullerene structure is similar to graphite, but a pentagonal ring may also exist in the fullerene structure. The most abundant form of fullerene is Buckminster fullerene (C 60), which has 60 carbon atoms arranged in a spherical structure.
- **Properties:** Like graphite, graphene conducts electricity well because it has delocalized electrons that are free to move across its surface. These

properties make graphene useful in electronics and for making composites. Graphene has a giant covalent structure, but fullerenes have large molecules.

- **Applications:** The discovery of C 60 has led to a paradigm shift in the understanding of graphite, in particular graphene sheets on a small scale. It is now known that the most stable form of a carbon aggregate, containing tens to several thousands of atoms, is a closed buckyball or nanotube.
- **Conclusion:** Graphite and fullerene are two forms of carbon that have unique structures and properties. They have a wide range of applications in electronics, composites, and other fields. Their discovery and study have led to a better understanding of the properties of carbon and its potential uses.

### **Nanomaterials: Introduction, Preparation, Characteristics, and Applications**

Nanomaterials are materials that have at least one dimension in the nanoscale range, typically between 1 and 100 nanometers. At this scale, unique optical, magnetic, electrical, and other properties emerge, which have great potential applications in electronics, medicine, and other fields.

Nanomaterials can be classified into nanostructured and nanophase/nanoparticle materials. The properties of nanomaterials can be tuned as desired via precisely controlling the size, shape, synthesis conditions, and appropriate functionalization.

Nanomaterials are being tested for applications in tissue engineering, drug delivery, and biosensors. Nanozymes are the artificial enzymes used for biosensing, bioimaging, and tumor detection. Nanomaterials are also utilized in various processes of manufacturing, health care, products as well as insulation, and so on.

The important characteristics of nanoparticles are facile manipulation and reversible assembly which allow for the possibility of incorporation of nanoparticles in electric, electronic or optical devices such as “bottom up” or “self-assembly” approaches are the benchmark of nanotechnology.

### **Nanomaterials: Carbon Nano Tubes (CNT)**

- Carbon nanotubes (CNTs) are cylindrical molecules made of carbon atoms, with diameters typically measuring in the nanometers.
- CNTs exhibit extraordinary strength and unique electrical properties, and are efficient conductors of heat.
- They have potential applications in nanotechnology, electronics, optics, and other fields of materials science.

- CNTs can be single-walled (SWCNTs) or multi-walled (MWCNTs).
- The structure of a CNT can be described by a vector (n,m) which represents the number of unit vectors along two directions in the honeycomb crystal lattice of graphene.
- CNTs can be synthesized using various methods, including chemical vapor deposition (CVD), arc discharge, and laser ablation.
- CNTs have potential applications in a wide range of fields, including electronics, energy storage, drug delivery, and structural materials.
- However, there are still challenges to be addressed in the large-scale production and application of CNTs, including the control of their structure and properties, and the potential environmental and health risks associated with their use.

### **Green Chemistry: Introduction, 12 principles and importance of green Synthesis, Green**

Green chemistry, also known as sustainable chemistry, is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal.

The 12 principles of green chemistry are:

1. Prevention: It is better to prevent waste than to treat or clean up waste after it has been created.
2. Atom Economy: Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. Less Hazardous Chemical Syntheses: Wherever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. Designing Safer Chemicals: Chemical products should be designed to affect their desired function while minimizing their toxicity.
5. Safer Solvents and Auxiliaries: The use of auxiliary substances (e.g., solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. Design for Energy Efficiency: Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
7. Use of Renewable Feedstocks: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. Reduce Derivatives: Unnecessary derivatization (use of blocking groups, protection/deprotection, temporary modification of physical/chemical processes) should be minimized or avoided if possible, because such steps require additional reagents and can generate waste.
9. Catalysis: Catalytic reagents (as selective as possible) are superior to

stoichiometric reagents.

10. Design for Degradation: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
11. Real-time analysis for Pollution Prevention: Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
12. Inherently Safer Chemistry for Accident Prevention: Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions, and fires.

The importance of green synthesis in green chemistry lies in its ability to minimize the use and generation of hazardous substances, reduce waste, and promote the use of renewable resources. Green synthesis methods aim to be more environmentally friendly, energy-efficient, and sustainable, while still producing high-quality chemical products. This approach can lead to significant benefits for both the environment and the economy. Green chemistry is an important field that has the potential to greatly improve the sustainability of the chemical industry and reduce its impact on the environment.

## **Chemicals, Synthesis of typical organic compounds by conventional and Green route (Adipic Acid)**

- Adipic acid is an organic compound that can be synthesized through both conventional and green routes.
- The conventional route for the synthesis of adipic acid involves the use of benzene as the starting material. This process requires high pressure and temperature .
- In contrast, the green route for the synthesis of adipic acid involves the use of glucose as the starting material. This process requires less temperature and pressure .
- Green chemistry techniques have been applied to synthesize both well-known chemical compounds by more sustainable routes and completely new materials .
- Green techniques in organic chemistry include reactions of C–H bond activation, fluororous, solid-supported, bio- and asymmetric catalysis and synthesis, use of water and other green solvents (in particular, ionic liquids (ILs)) or without any solvent, microwave-, ultrasound- and ultraviolet (UV)-assisted reactions, and use of flow reactors .
- The ideology of Green Chemistry calls for the development of new chemical reactivities and reaction conditions that can potentially provide benefits for chemical syntheses in terms of resource and energy efficiency, product

selectivity, operational simplicity, and health and environmental safety .

## Acid and Paracetamol

- Mefenamic Acid + Paracetamol is a combination of two medicines: Mefenamic Acid and Paracetamol. It works by blocking the release of certain chemical messengers that cause fever, pain and inflammation (redness and swelling).
- Paracetamol may be made by acetylation of para-aminophenol (obtained by reduction of para-nitrophenol) with acetic acid or acetic anhydride.
- Paracetamol is used as an analgesic and antipyretic drug. It is the preferred alternative analgesic-antipyretic to aspirin (acetylsalicylic acid), particularly in patients with coagulation disorders, individuals with a history of peptic ulcer or who cannot tolerate aspirin, as well as in children.

## Environmental impact of Green chemistry on society

- Green chemicals either degrade to innocuous products or are recovered for further use. Plants and animals suffer less harm from toxic chemicals in the environment. Lower potential for global warming, ozone depletion, and smog formation.
- Scientists and chemists can significantly minimize the risk to environment and health of human by the help of all the valuable ideology of green chemistry.
- Green chemistry is the design of high-performing, cost-effective technology that is safer for the environment and human health.
- In the Global Chemicals Outlook II it was concluded that the global goal to minimize adverse impacts of chemicals and waste will not be achieved by 2020. More urgent and ambitious worldwide action by stakeholders is required.

### Unit-2: 8

1. Point 1
2. Point 2
3. Point 3
4. Point 4
5. Point 5
6. Point 6
7. Point 7
8. Point 8

## Spectroscopic Techniques and Applications: Elementary idea and simple applications of

Spectroscopy is the study of the interaction between matter and electromagnetic radiation. It is a powerful tool for the analysis of the structure and composition of materials. Some of the elementary ideas and simple applications of spectroscopy are:

1. **Identification of elements and compounds:** Spectroscopy can be used to identify the presence of specific elements or compounds in a sample by analyzing the characteristic absorption or emission spectra of the sample.
2. **Determination of molecular structure:** Spectroscopy can provide information about the bonding and geometry of molecules by analyzing the vibrational and rotational spectra of the molecules.
3. **Quantitative analysis:** Spectroscopy can be used to determine the concentration of a specific compound in a sample by measuring the intensity of its characteristic absorption or emission lines.
4. **Remote sensing:** Spectroscopy can be used to remotely analyze the composition of materials, such as the atmosphere of a planet or the surface of a distant object, by analyzing the absorption or emission spectra of the light reflected or emitted by the material.

These are just a few examples of the many applications of spectroscopy. Spectroscopy is a versatile and powerful tool that is widely used in many fields, including chemistry, physics, biology, and environmental science.

## UV, IR and NMR, Numerical problems

UV, IR and NMR are three different types of spectroscopy techniques used to analyze the structure of molecules.

- **UV spectroscopy** uses ultraviolet and visible light to analyze the electronic structure of molecules. It is commonly used to determine the concentration of a substance in a solution.
- **IR spectroscopy** uses infrared radiation to analyze the vibrational structure of molecules. It is commonly used to identify functional groups in organic molecules.
- **NMR spectroscopy** uses radio waves to analyze the magnetic properties of atomic nuclei. It is commonly used to determine the structure of organic molecules.

Numerical problems in these techniques involve using the principles of spectroscopy to solve problems related to the structure and properties of molecules. These problems may involve calculating the concentration of a substance in a so-



lution, identifying functional groups in a molecule, or determining the structure of a molecule.

Here are some examples of numerical problems in UV, IR and NMR spectroscopy:

1. A solution of compound X has an absorbance of 0.75 at a wavelength of 260 nm. The molar absorptivity of compound X at this wavelength is 15,000 L/mol/cm. Calculate the concentration of compound X in the solution.
2. An IR spectrum of an unknown compound shows a strong absorption at 1710 cm<sup>-1</sup>. What functional group is likely present in the molecule?
3. An NMR spectrum of an unknown compound shows a singlet at 3.2 ppm with an integration of 3. What is the likely structure of the molecule?

These are just a few examples of the types of numerical problems that can be encountered in UV, IR and NMR spectroscopy. To solve these problems, it is important to have a strong understanding of the principles of spectroscopy and the properties of molecules.

### **Stereochemistry: Optical isomerism in compounds without chiral carbon, Geometrical**

- Optical isomerism occurs when a compound has similar bonds but a different spatial arrangement of atoms, resulting in non-superimposable mirror images, known as optical isomers or enantiomers.
- Enantiomers differ from one another in their optical activity.
- Stereochemistry can explain the organization of stereoisomers.
- Geometrical stereoisomers are a common occurrence in heteroleptic compounds. This type of isomerism occurs due to the different geometric configurations for the chemicals.
- Optical isomerism occurs in chiral particles in solution, which appear to be mirror images of one another.
- Optical isomerism is one form of stereoisomerism; geometric isomers are a second type.
- Optical isomers are named as such because of their effect on plane polarized light. Simple substances that show optical isomerism exist as two isomers known as enantiomers.

## **Isomerism and Chiral Drugs**

- Isomerism is the process of drug degradation where a drug gets converted into its optical or geometric isomers. This process is known as isomerization.
- Stereoisomers are isomeric pairs with identical atomic and molecular formulas but different three-dimensional arrangements.

- Many drugs available on the market exhibit optical isomerism. A molecule that cannot be superimposed on its mirror image is called chiral, and each of these images is called an enantiomer. Enantiomers are isomers that are mirror images of each other.
- The stereoisomeric composition of a drug with a chiral center should be known and the quantitative isomeric composition of the material used in pharmacologic, toxicologic, and clinical studies should be determined.
- The importance of chiral drugs in the drug development space cannot be understated. In pharmaceutical industries, 56% of the drugs currently in use are chiral molecules and 88% of the last ones are marketed as racemates (or racemic mixtures), consisting of an equimolar mixture of two enantiomers.
- The word chiral describes the three-dimensional architecture of the molecule and does not reveal the stereochemical composition. Hence “chiral drug” does not say whether the drug is racemic (racemic drug), single enantiomer (chiral specific drug) or some other combination of stereoisomers.

### Unit-3: 8

1. I'm sorry, but I'm not sure what you're referring to with “Unit-3: 8”. Could you please provide more context or information about the topic you would like me to write about?
2. Is this a chapter or section in a specific book or course?

### Electrochemistry and Batteries: Basic concepts of electrochemistry

Electrochemistry is the branch of chemistry that deals with the relationship between electrical energy and chemical reactions. It involves the study of the movement of electrons in chemical reactions, particularly in redox reactions where the oxidation state of atoms changes.

Some basic concepts of electrochemistry include:

1. **Electrochemical cell:** An electrochemical cell is a device that converts chemical energy into electrical energy or vice versa. It consists of two electrodes, an anode and a cathode, immersed in an electrolyte solution.
2. **Anode and Cathode:** The anode is the electrode where oxidation occurs, and the cathode is the electrode where reduction occurs. Electrons flow from the anode to the cathode.
3. **Electrolyte:** An electrolyte is a substance that conducts electricity when dissolved in water. It contains ions that can move freely to carry electric charge.
4. **Redox reaction:** A redox reaction is a chemical reaction in which the oxidation state of one or more atoms is changed. It involves the transfer of electrons between chemical species.

5. **Oxidation and Reduction:** Oxidation is the loss of electrons, while reduction is the gain of electrons. In a redox reaction, one species is oxidized, and another is reduced.
6. **Battery:** A battery is a device that converts chemical energy into electrical energy. It consists of one or more electrochemical cells connected in series or parallel.

### Batteries; Classification and applications of Primary Cells (Dry Cell) and Secondary Cells

A battery is a combination of many cells, which can be divided into two types: primary and secondary cells .

- **Primary Cells:** A primary cell is a battery that is designed to be used only one time. It creates an electric current through an irreversible electrochemical reaction occurring in the cell. When a primary cell is discharged, it is discarded, and not recharged again like the secondary cell “rechargeable battery” . Dry cells and (most) alkaline batteries are examples of primary batteries .
- **Secondary Cells:** The second type of battery is rechargeable and is called a secondary battery. Examples of secondary batteries include nickel-cadmium (NiCd), lead acid, and lithium-ion batteries .

## Lead Acid Battery

- The lead-acid battery is a type of rechargeable battery first invented in 1859 by French physicist Gaston Planté .
- It is the first type of rechargeable battery ever created .
- Compared to modern rechargeable batteries, lead-acid batteries have relatively low energy density .
- A lead-acid battery consists of a negative electrode made of spongy or porous lead .
- The positive electrode consists of lead oxide .
- The lead-acid battery consists of two electrodes submerged in an electrolyte of sulfuric acid .
- Lead-acid batteries are classified into two types: flooded and valve-regulated .
- Lead-acid batteries are currently used in uninterrupted power modules, electric grid, and automotive applications .
- A lead-acid battery system may cost hundreds or thousands of dollars less than a similarly-sized lithium-ion setup .

## Corrosion

- Corrosion is the process of deterioration of metal caused by the action of chemicals or electro-chemicals present in the surrounding atmosphere.
- Corrosion is a major problem especially in the construction industry where various metals are used for structural purposes.
- Corrosion is the chemical or electrochemical destruction of metals and alloys in the presence of an environment to form undesirable compounds.
- Corrosion processes are the reactions of metals with environmental organisms.
- It is the process of a body gradually deteriorating due to the attack of air gases on the metal's surface.
- Corrosion is the gradual deterioration of metals caused by the action of air, moisture, or a chemical reaction (such as an acid) on their surface.
- Rusting of iron, or the forming of a brown flaky material on iron objects when exposed to moist air, is the most common example of metal corrosion.
- The different forms of corrosion can be categorized into three groups by the way in which it is identified: Group 1: Corrosion readily identifiable by ordinary visual examination Uniform Pitting Crevice Fillform Pack Rust Galvanic Lamellar Group 2: Corrosion requiring supplementary means of examination Erosion Cavitation Fretting Intergranular.

## Prevention and Control of Corrosion

Corrosion prevention and control (CPC) is a crucial aspect of maintaining physical assets, ranging from aircraft, ships, ground combat vehicles, and materiel, as well as wharves, buildings, and other infrastructure. These assets are subject to degradation due to corrosion, and CPC helps to reduce the effect of corrosion on sustainment costs while also improving materiel readiness.

CPC entails the characteristics of a system design to preclude or reduce corrosion, materials selection, non-destructive inspections for corrosion detection, coatings, finishes, cleaning materials and washings, repairs, and other maintenance activities .

The DoD Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment (MS&E) (i.e., CPC Planning Guidebook) helps Program Managers (PMs), Systems Engineers, Product Support Managers and other program staff develop and execute a comprehensive CPC approach.

## Issues in Specific Industries: Power Generation and Chemical

### Power Generation

1. **Environmental Impact:** Power generation, particularly from fossil fuels, contributes to air and water pollution, greenhouse gas emissions, and habitat destruction.
2. **Reliability and Resilience:** The power grid is vulnerable to natural disasters, cyber attacks, and physical attacks, which can disrupt the supply of electricity.
3. **Aging Infrastructure:** Much of the power generation infrastructure is aging and in need of repair or replacement, which can be costly and time-consuming.
4. **Regulatory Challenges:** The power generation industry is subject to a complex web of regulations at the local, state, and federal levels, which can create challenges for companies operating in this space.

### Chemical

1. **Environmental Impact:** The chemical industry can have a significant impact on the environment, including air and water pollution, and the release of hazardous waste.
2. **Worker Safety:** The chemical industry can be dangerous for workers, with risks including exposure to hazardous chemicals, fires, and explosions.
3. **Regulatory Compliance:** The chemical industry is subject to a wide range of regulations, including those related to environmental protection, worker safety, and the transportation of hazardous materials.
4. **Supply Chain Disruptions:** The chemical industry relies on a complex global supply chain, which can be disrupted by natural disasters, trade disputes, and other factors.

## Processing Industry, Oil & Gas Industry, and Pulp & Paper Industries

### Processing Industry

- The processing industry involves the transformation of raw materials into finished products.
- This industry includes a wide range of sectors, such as food processing, chemical processing, and mineral processing.
- The processing industry plays a crucial role in the economy, as it adds value to raw materials and creates employment opportunities.

## Oil & Gas Industry

- The oil and gas industry involves the exploration, extraction, refining, and distribution of oil and gas.
- This industry is vital for the global economy, as it provides energy for transportation, heating, and electricity generation.
- The oil and gas industry is facing challenges, such as the need to reduce greenhouse gas emissions and to find new sources of oil and gas.

## Pulp & Paper Industries

- The pulp and paper industries involve the production of pulp and paper from wood or other raw materials.
- This industry is important for the production of paper products, such as newspapers, books, and packaging materials.
- The pulp and paper industries are facing challenges, such as the need to reduce their environmental impact and to find new sources of raw materials.

## Chemistry of Engineering Materials:

1. Engineering materials are materials that are used in the construction of machines, structures, and other technical systems.
2. The chemistry of engineering materials is concerned with the study of the chemical properties and behavior of these materials.
3. Some common engineering materials include metals, ceramics, polymers, and composites.
4. The chemical properties of these materials, such as their reactivity, corrosion resistance, and thermal stability, are important factors in their selection and use in engineering applications.
5. The study of the chemistry of engineering materials also involves the development of new materials with improved properties, as well as the modification of existing materials to enhance their performance.
6. Understanding the chemistry of engineering materials is essential for the design and development of new technologies and systems.

## Cement; Constituents, manufacturing, hardening and setting, deterioration of cement, Plaster

1. **Constituents:** Cement is a binding material that sets, hardens, and adheres to other materials to bind them together. It is manufactured through a closely controlled chemical combination of calcium, silicon, aluminum, iron, and other ingredients.
2. **Manufacturing:** Cements of this kind are finely ground powders that, when mixed with water, set to a hard mass. Setting and hardening result from hydration, which is a chemical combination of the cement compounds

with water that yields submicroscopic crystals or a gel-like material with a high surface area.

3. **Hardening and Setting:** Hydration, setting, and hardening of Portland cement are constrained by C3S, C3A. And all the admixtures that affect the hydration of C3S, C3A can change the performance of the hydration, the setting, and hardening of Portland cement.
4. **Deterioration of Cement:** Set cement and concrete can suffer deterioration from attack by some natural or artificial chemical agents. The alumina compound is the most vulnerable to chemical attack in soils containing sulfate salts or in seawater, while the iron compound and the two calcium silicates are more resistant.
5. **Plaster:** Plaster is a building material used for the protective or decorative coating of walls and ceilings and for molding and casting decorative elements. It is made by heating gypsum to around 150°C. When the dry plaster powder is mixed with water, it re-forms into gypsum.

## Paris Population

- Paris is the capital and most populous city of France.
- As of January 1, 2019, the population of Paris was 2,165,423 residents within its administrative city limits .
- Paris is the fourth-most populated city in the European Union as well as the 30th most densely populated city in the world in 2022 .
- The city is surrounded by the Paris unité urbaine, or urban area, the most populous urban area in the European Union .
- The suburban area of Paris is home to 11 million people, while the city proper has a population of 2.1 million .

## Unit-4: 8

1. I'm sorry, but I need more information to provide a response. Can you please clarify what you mean by "Unit-4: 8"?
2. Is this a chapter or section in a specific book or course?

## Water Technology

Water is a vital resource for life and industrial processes. It is obtained from various sources and may contain impurities that affect its usability. In this section, we will discuss the sources and impurities of water, hardness of water, and boiler troubles.

### Sources of Water

Water can be obtained from various sources, including:

1. Surface water: This includes water from rivers, lakes, and reservoirs.

2. Groundwater: This is water that is found underground in aquifers.
3. Rainwater: This is water that is collected from precipitation.
4. Seawater: This is water from oceans and seas that can be desalinated to make it usable.

### **Impurities in Water**

Water from different sources may contain various impurities, including:

1. Suspended solids: These are particles that are suspended in water and can make it appear cloudy or turbid.
2. Dissolved solids: These are minerals and salts that are dissolved in water and can affect its taste and usability.
3. Organic matter: This includes plant and animal matter that can decompose in water and affect its quality.
4. Microorganisms: These include bacteria, viruses, and other pathogens that can cause illness.

### **Hardness of Water**

Hardness of water refers to the presence of dissolved minerals, particularly calcium and magnesium, in water. Hard water can cause scaling in pipes and boilers, and can reduce the effectiveness of soaps and detergents.

### **Boiler Troubles**

Boiler troubles refer to problems that can occur in boilers due to the presence of impurities in water. These include:

1. Scaling: This is the buildup of minerals on the inside of boiler tubes, which can reduce heat transfer and efficiency.
2. Corrosion: This is the deterioration of metal due to chemical reactions with water and its impurities.
3. Priming and foaming: This refers to the carryover of water and impurities into the steam, which can cause damage to equipment and reduce efficiency.

In summary, water is obtained from various sources and may contain impurities that affect its usability. Hardness of water and boiler troubles are important considerations in industrial processes that use water. It is important to properly treat water to remove impurities and prevent problems.

## **Techniques for Water Softening**

Water softening is the process of removing impurities such as calcium and magnesium ions from water to make it more suitable for use. There are several techniques for water softening, including Lime-Soda, Zeolite, Ion Exchange, and Reverse Osmosis.



1. **Lime-Soda:** The lime-soda method of water softening involves the use of chemicals such as ammonia, borax, calcium hydroxide (slaked lime), or trisodium phosphate, usually in conjunction with sodium carbonate (soda ash). This method must be followed by sedimentation and filtration in order to remove the precipitates.
2. **Zeolite:** The zeolite process for water softening has become a commercial success because zeolite can be easily regenerated. When hard water containing calcium and magnesium ions is passed through a bed of sodium zeolite, the sodium ions are replaced by the calcium and magnesium ions.
3. **Ion Exchange:** Ion exchange softening is applied to water high in non-carbonated hardness and the total hardness does not exceed 350 mg/L. This method involves the use of ion exchange resins to remove calcium and magnesium ions from water.
4. **Reverse Osmosis:** Reverse osmosis softening involves the use of a semipermeable membrane to remove impurities from water. Water is forced through the membrane, leaving the impurities behind.

These are some of the common techniques used for water softening. Each method has its own advantages and disadvantages and the choice of method depends on the specific requirements of the application.

## Process of Determination of Hardness and Alkalinity

Hardness and alkalinity are important parameters in water quality analysis. Hardness refers to the presence of dissolved calcium and magnesium ions in water, while alkalinity refers to the water's ability to neutralize acids.

### Determination of Hardness

1. One of the most common methods for determining the hardness of water is by using a titration method with ethylenediaminetetraacetic acid (EDTA) as the titrant.
2. A buffer solution is added to the water sample to adjust the pH to around 10.
3. A few drops of an indicator, such as Eriochrome Black T, are added to the sample. The indicator will change color from red to blue when all the calcium and magnesium ions have been complexed by the EDTA.
4. The EDTA solution is then added to the sample until the color change occurs, indicating the end point of the titration.
5. The volume of EDTA used is then used to calculate the hardness of the water sample.

## Determination of Alkalinity

1. Alkalinity is determined by titrating a water sample with a strong acid, such as sulfuric acid or hydrochloric acid, to a pH of around 4.5.
2. An indicator, such as phenolphthalein or bromocresol green, is used to indicate the end point of the titration.
3. The volume of acid used is then used to calculate the alkalinity of the water sample.

## Numerical Problems

1. A water sample has a hardness of 200 ppm as  $\text{CaCO}_3$ . What is the hardness in mg/L as  $\text{Ca}^{2+}$ ?
2. A water sample has an alkalinity of 150 ppm as  $\text{CaCO}_3$ . What is the alkalinity in meq/L?

## Fuels and Combustion

- **Definition:** A fuel is any material that can be burned to release energy in the form of heat. Combustion is the process of burning a fuel in the presence of oxygen to produce heat and light.
- **Classification:** Fuels can be classified into three main categories: solid, liquid, and gaseous. Solid fuels include coal, wood, and charcoal. Liquid fuels include petroleum, diesel, and kerosene. Gaseous fuels include natural gas, propane, and hydrogen.
- **Characteristics of a good fuel:** A good fuel should have a high calorific value, which means it should release a large amount of heat when burned. It should also be easy to ignite, burn cleanly without producing harmful emissions, and be readily available at a reasonable cost.
- **Calorific value:** The calorific value of a fuel is the amount of heat energy released when a unit quantity of the fuel is burned completely. It is usually measured in units of energy per unit mass, such as joules per gram or calories per gram. The higher the calorific value of a fuel, the more heat it will release when burned.

## Values, Gross & Net Calorific Value, Determination of Calorific Value by Bomb Calorimeter

### Values

- Values are principles or standards of behavior that are considered important or worthwhile.
- They can be personal, cultural, or societal and can vary between individuals and groups.

- Values can influence decision-making and behavior.

## **Gross & Net Calorific Value**

- Calorific value is the amount of heat energy released when a substance is burned.
- Gross calorific value (GCV) is the total amount of heat released when a substance is burned, including the heat of condensation of the water vapor produced.
- Net calorific value (NCV) is the amount of heat released when a substance is burned, excluding the heat of condensation of the water vapor produced.
- NCV is typically lower than GCV because it does not include the heat of condensation of the water vapor produced.

## **Determination of Calorific Value by Bomb Calorimeter**

- A bomb calorimeter is a device used to measure the calorific value of a substance.
- The substance is placed in a sealed container (the “bomb”) and burned in an oxygen-rich environment.
- The heat released by the combustion is absorbed by the surrounding water and the temperature rise is measured.
- The calorific value can be calculated from the temperature rise and the heat capacity of the calorimeter.

## **Theoretical calculation of calorific value by Dulong’s method, Ranking of Coal, Analysis of coal**

### **Theoretical calculation of calorific value by Dulong’s method**

- The various formulae for calculating calorific values for coals from ultimate analyses depend essentially on a proposition due to Dulong.
- The heat of combustion of an organic compound is nearly equal to the heats of combustion of the elements in it, multiplied by their percentage content in the compound in question .

### **Ranking of Coal**

- Coal is ranked based on its degree of transformation from the original organic material into carbon.
- The higher the carbon content, the higher the rank of the coal.
- The ranking of coal, from lowest to highest, is lignite, sub-bituminous, bituminous, and anthracite.

## Analysis of coal

- Coal analysis is the process of determining the composition and properties of coal.
- It is used to determine the quality of the coal and its suitability for specific applications.
- Coal analysis can include measurements of the coal's moisture, volatile matter, ash, sulfur, and carbon content, as well as its heating value and other properties.

## Proximate and Ultimate Analysis Method

Proximate analysis is a method used to determine the composition of coal, biomass, and other solid fuels. It measures the following parameters: - Moisture content - Volatile matter - Fixed carbon - Ash content

Ultimate analysis, on the other hand, determines the elemental composition of the fuel, including: - Carbon - Hydrogen - Nitrogen - Sulfur - Oxygen

## Numerical Problems

Numerical problems related to proximate and ultimate analysis involve calculating the composition of the fuel based on the given data. For example, if the moisture content, volatile matter, fixed carbon, and ash content of a coal sample are given, the heating value of the coal can be calculated.

## Chemistry of Biogas

Biogas is a renewable energy source produced by the anaerobic digestion of organic matter. The main components of biogas are methane ( $\text{CH}_4$ ) and carbon dioxide ( $\text{CO}_2$ ). The anaerobic digestion process involves the breakdown of organic matter by microorganisms in the absence of oxygen. The process can be divided into four stages: 1. Hydrolysis 2. Acidogenesis 3. Acetogenesis 4. Methanogenesis

During hydrolysis, complex organic compounds are broken down into simpler compounds. In acidogenesis, these simple compounds are further broken down into volatile fatty acids and other byproducts. During acetogenesis, the volatile fatty acids are converted into acetic acid, hydrogen, and carbon dioxide. Finally, during methanogenesis, methane is produced from acetic acid, hydrogen, and carbon dioxide.

## Production from Organic Waste Materials and their Environmental Impact on Society

Organic waste materials can be used for production in various ways. One of the most common methods is composting, which is the process of decomposing organic matter to produce a nutrient-rich soil amendment. Composting can significantly reduce methane emissions, a potent greenhouse gas, that is generated when organic waste is disposed of in landfills. Compost can also reduce or eliminate the need for chemical fertilizers.

Organic waste feedstocks of aquatic, agricultural, forestry, and terrestrial animal origin are suitable for valorization. Valorization is the process of converting waste materials into valuable products, such as biofuels, fertilizers, and animal feed.

However, the disposal of organic waste in landfills can have a negative impact on the environment. Anaerobic decomposition of organic materials in landfills produces methane, a greenhouse gas with a global warming potential approximately 85 times higher than carbon dioxide over a 20-year time period. Landfills are one of the top emitters of man-made methane emissions in the United States.

In conclusion, the production from organic waste materials can have a positive impact on the environment by reducing greenhouse gas emissions and the need for chemical fertilizers. However, the disposal of organic waste in landfills can have a negative impact on the environment by producing methane, a potent greenhouse gas.

### Unit-5: 8

1. I'm sorry, but I need more information to provide a detailed response on the topic "Unit-5: 8". Could you please clarify what subject or context this unit refers to?
2. Once I have more information, I can provide a formal and informative response in markdown format, without the use of external links or emojis, and written in a way that is suitable for studying and exam preparation.

### Materials Chemistry:

Materials Chemistry is a branch of chemistry that deals with the study of the synthesis, structure, properties, and performance of materials. These materials can range from simple substances to complex, multi-component systems, and can be organic, inorganic, or a combination of both.

Some key points to consider when studying Materials Chemistry include:

1. The synthesis of materials involves the creation of new materials through chemical reactions or other processes. This can include the use of traditional chemical synthesis techniques, as well as newer methods such as

self-assembly or the use of templates.

2. The structure of materials refers to the arrangement of atoms, molecules, or ions within the material. This can include the study of crystal structures, as well as the arrangement of molecules in amorphous materials.
3. The properties of materials are determined by their structure and composition. These can include physical properties such as melting point, density, and conductivity, as well as chemical properties such as reactivity and stability.
4. The performance of materials refers to how well they function in their intended applications. This can include factors such as durability, efficiency, and effectiveness.

Materials Chemistry is a broad and interdisciplinary field, with applications in many areas including energy, electronics, medicine, and the environment. It is a rapidly evolving field, with new discoveries and developments being made on a regular basis.

### **Polymers; Classification, Polymerization processes, Thermosetting and Thermoplastic**

Polymers are large molecules composed of repeating subunits called monomers. They can be classified based on their origin, structure, and properties.

#### **Classification of Polymers**

- **Based on origin:** Polymers can be classified as natural or synthetic. Natural polymers are found in nature, such as cellulose, starch, and proteins. Synthetic polymers are man-made, such as plastics, synthetic fibers, and rubber.
- **Based on structure:** Polymers can be classified as linear, branched, or cross-linked. Linear polymers have a simple, long-chain structure. Branched polymers have side chains attached to the main chain. Cross-linked polymers have covalent bonds between the chains.
- **Based on properties:** Polymers can be classified as thermosetting or thermoplastic. Thermosetting polymers are rigid and cannot be remolded once set. Thermoplastic polymers can be melted and remolded multiple times.

#### **Polymerization Processes**

Polymerization is the process of forming a polymer by combining monomers. There are two main types of polymerization processes: addition and condensation. - **Addition polymerization:** In this process, monomers with unsaturated bonds (double or triple bonds) react to form a polymer. The reaction is initiated by a free radical, which attacks the unsaturated bond and creates a new

free radical. This new free radical then attacks another monomer, and the process continues until the polymer is formed. - **Condensation polymerization:** In this process, monomers with functional groups react to form a polymer. The reaction involves the elimination of a small molecule, such as water or methanol. The polymer is formed by the repeated condensation of monomers.

### Thermosetting and Thermoplastic Polymers

- **Thermosetting polymers:** These polymers are rigid and cannot be remolded once set. They are formed by cross-linking, which creates a three-dimensional network of covalent bonds. Examples of thermosetting polymers include epoxy, melamine, and phenolic resins.
- **Thermoplastic polymers:** These polymers can be melted and remolded multiple times. They have a linear or branched structure, with weak intermolecular forces between the chains. Examples of thermoplastic polymers include polyethylene, polypropylene, and polyvinyl chloride.

## Polymers, Polymer Blends and Composites, Conducting and Biodegradable polymers

### Polymers

- Polymers are large molecules made up of repeating subunits called monomers.
- Polymers can be natural or synthetic.
- Examples of natural polymers include proteins, DNA, and cellulose.
- Examples of synthetic polymers include plastics, such as polyethylene and polypropylene.

### Polymer Blends and Composites

- Polymer blends are mixtures of two or more polymers.
- Polymer composites are materials made by combining a polymer with another material, such as a fiber or a filler.
- Polymer blends and composites can have improved properties compared to the individual components.
- Examples of polymer composites include fiber-reinforced plastics and wood-plastic composites.

### Conducting Polymers

- Conducting polymers are a class of polymers that can conduct electricity.
- Conducting polymers can be used in applications such as batteries, sensors, and displays.
- Examples of conducting polymers include polyaniline, polypyrrole, and polythiophene.

## Biodegradable Polymers

- Biodegradable polymers are polymers that can be broken down by microorganisms into natural substances.
- Biodegradable polymers can be used in applications such as compostable packaging and medical implants.
- Examples of biodegradable polymers include polylactic acid, polyhydroxyalkanoates, and starch-based polymers.

## Preparation, properties, industrial applications of Teflon, Lucite, Bakelite, Kelvar, Dacron

### Teflon

Teflon is a brand name for the chemical compound polytetrafluoroethylene (PTFE). It is a thermally stable material that can withstand high temperatures. Its chemical properties make it ideal for various real-world applications, including electrical appliances, manufacturing, and the textile industry, where it is used in the production of many clothing items .

#### Properties of Teflon

- It is a white solid compound at room temperature.
- Its density is about 2200 kg/m<sup>3</sup> or 2.2 g/cm<sup>3</sup>.
- Its melting point is 600 K.
- It is a chemically resistant compound, with the only chemicals that can affect it being alkali metals.
- It shows good resistance towards heat and low temperature.
- It has a low water absorption capacity .

### Bakelite

Bakelite is a type of plastic that is made from phenol and formaldehyde. It is prepared by removing a water molecule during the polymerization process of hydroxy benzyl alcohol, formed from phenol and formaldehyde in an acid medium with HCl or ZnCl<sub>2</sub> .

#### Properties of Bakelite

- It can be formed quickly.
- Very smooth molding can be obtained from this polymer.
- Bakelite moldings are heat resistant and resistant to scratches .

#### Industrial Applications of Bakelite

Bakelite is a good insulator and is used in the manufacture of non-conducting parts of radio and electric devices such as sockets, wire insulation, switches,



and automobile distribution caps. It is also used in the manufacture of clocks, buttons, washing machines, toys, kitchenware, and other items .

### Lucite

I'm sorry, but I couldn't find any information on Lucite in my search results.

### Kelvar

I'm sorry, but I couldn't find any information on Kelvar in my search results.

### Dacron

I'm sorry, but I couldn't find any information on Dacron in my search results.

## Thiokol, Nylon, Buna-N and Buna-S and their environmental impact on society, Speciality

- **Nylon** is the world's first entirely synthetic polymer fiber, introduced by the DuPont company in 1938. It is known for its strength, durability, and flexibility.
- **Buna-N**, also known as **nitrile rubber**, is a synthetic rubber derived from acrylonitrile (ACN) and butadiene. It is resistant to oil, fuel, and other chemicals.
- **Buna-S**, now known as **styrene butadiene rubber (SBR)**, was being produced in large quantities in Germany by 1935.
- **Thiokol** is not mentioned in the search results.

## Polymers

- Polymers range from familiar synthetic plastics such as polystyrene to natural biopolymers such as DNA and proteins that are fundamental to biological structure and function.
- Polymers, both natural and synthetic, are created via polymerization of many small molecules, known as monomers.
- A polymer is a substance composed of macromolecules.
- A macromolecule is a molecule of high relative molecular mass, the structure of which essentially comprises the multiple repetition of units derived, actually or conceptually, from molecules of low relative molecular mass.
- Polymers make up many of the materials in living organisms, including, for example, proteins, cellulose, and nucleic acids.

## Organometallic Compounds: General methods of preparation and applications of

Organometallic compounds are chemical compounds that contain at least one bond between a carbon atom of an organic molecule and a metal. These compounds have a wide range of applications in various fields, including catalysis, medicine, and materials science.

### General methods of preparation:

1. **Direct reaction of a metal with an organic compound:** This method involves the direct reaction of a metal with an organic compound, such as an alkane, alkene, or alkyne, to form an organometallic compound. For example, the reaction of magnesium with an alkyl halide produces a Grignard reagent, which is an organomagnesium compound.
2. **Reaction of a metal with an organic halide:** This method involves the reaction of a metal with an organic halide, such as an alkyl or aryl halide, to form an organometallic compound. For example, the reaction of lithium with an alkyl halide produces an organolithium compound.
3. **Transmetalation:** This method involves the transfer of an organic group from one metal to another. For example, the reaction of an organolithium compound with a transition metal halide produces an organotransition metal compound.

### Applications of organometallic compounds:

1. **Catalysis:** Organometallic compounds are widely used as catalysts in various chemical reactions, such as hydrogenation, hydroformylation, and polymerization.
2. **Medicine:** Organometallic compounds have been used in the development of various drugs, such as anti-cancer agents and anti-inflammatory agents.
3. **Materials science:** Organometallic compounds have been used in the development of various materials, such as conductive polymers and light-emitting diodes.

## Organometallic Compounds ( $\text{RMgX}$ and $\text{LiAlH}_4$ )

Organometallic compounds are compounds that contain a metal-carbon bond.  $\text{RMgX}$ , where X is a halide, is known as a Grignard reagent. Grignard reagents are named after Victor Grignard. They are usually written as  $\text{R-Mg-X}$ , but in fact, the magnesium(II) center is tetrahedral when dissolved in Lewis basic solvents.

Grignard reagents are important due to their use in organic synthesis and as synthons for a range of organometallic compounds. They react with organic carbonyls (aldehydes, ketones, and esters) to yield the alcohol on hydrolysis. This synthetic route is useful for the formation of primary, secondary, and terminal alcohols.

$\text{LiAlH}_4$  is another important organometallic compound. It is a reducing agent commonly used in organic synthesis. It can reduce aldehydes, ketones, carboxylic acids, and esters to their corresponding alcohols.

### Course Outcomes:

- Define and explain the key concepts and principles of the subject.
- Demonstrate an understanding of the subject's theories and their applications.
- Analyze and evaluate information, arguments, and evidence related to the subject.
- Apply the subject's concepts and principles to solve problems and make decisions.
- Communicate effectively in written and oral forms about the subject.
- Work effectively in a team to achieve common goals related to the subject.
- Demonstrate ethical and professional behavior in the context of the subject.
- Engage in lifelong learning to stay current in the subject.

### Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the knowledge and skills acquired in the course to solve problems.
3. Communicate effectively in both written and oral forms.
4. Work effectively in a team.
5. Demonstrate critical thinking and analytical skills.
6. Demonstrate ethical and professional behavior.
7. Demonstrate the ability to learn independently and continuously.
8. Demonstrate the ability to adapt to new situations and technologies.
9. Demonstrate the ability to integrate knowledge from different disciplines.
10. Demonstrate the ability to apply knowledge in practical situations.

### Units Course Outcomes Bloom's

1. **Units:** Units are a standard of measurement used to express physical quantities. They provide a common language for scientists and engineers to communicate their findings and ideas.
2. **Course Outcomes:** Course outcomes are the specific knowledge, skills, and abilities that students are expected to achieve by the end of a course.

They are used to assess the effectiveness of the course and to guide future improvements.

3. **Bloom's Taxonomy:** Bloom's Taxonomy is a framework for categorizing educational goals and objectives. It is used to design learning activities and assessments that promote higher-order thinking skills. The taxonomy consists of six levels: remembering, understanding, applying, analyzing, evaluating, and creating.

#### Level

- A level is a tool used to determine if a surface is horizontal or vertical.
- It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
- When the level is placed on a surface, the bubble will move to the highest point of the tube, indicating if the surface is level or not.
- There are different types of levels, including spirit levels, laser levels, and digital levels.
- Levels are commonly used in construction, carpentry, and surveying, among other fields.
- It is important to use a level when installing objects such as shelves or cabinets to ensure they are straight and level.
- Levels can also be used to check the slope of a surface, such as a driveway or a garden bed, to ensure proper drainage.
- Using a level can help improve the accuracy and quality of a project.

#### CO-1 Get an understanding of the theoretical principles of chemistry of molecular structure, bonding and

1. **Molecular structure** refers to the three-dimensional arrangement of atoms within a molecule. It is determined by the type of atoms, the number of atoms, and the type of chemical bonds between the atoms.
2. **Chemical bonding** is the process by which atoms combine to form molecules. There are several types of chemical bonds, including covalent bonds, ionic bonds, and metallic bonds.
3. **Covalent bonds** are formed when atoms share electrons. This type of bond is common in organic molecules, such as those found in living organisms.
4. **Ionic bonds** are formed when one atom transfers an electron to another atom. This type of bond is common in salts, such as sodium chloride (NaCl).
5. **Metallic bonds** are formed when electrons are shared among many atoms. This type of bond is common in metals, such as copper and iron.
6. The **theoretical principles** of chemistry help us understand how and why atoms combine to form molecules, and how the properties of molecules are determined by their molecular structure and bonding.

## Properties, Chemistry of Advanced Materials

### Liquid Crystals

- Liquid crystals are a state of matter that has properties between those of a conventional liquid and those of a solid crystal.
- They can flow like a liquid, but have the long-range order of a solid.
- Liquid crystals are used in displays, such as LCD screens, because they can be manipulated to control the passage of light.

### Nanomaterials

- Nanomaterials are materials with at least one dimension in the nanometer scale (1-100 nm).
- They have unique physical and chemical properties due to their small size and high surface area.
- Nanomaterials are used in a wide range of applications, including electronics, medicine, and energy production.

### Graphite

- Graphite is a form of carbon in which the carbon atoms are arranged in layers.
- It is a good conductor of electricity and heat, and is used in batteries, lubricants, and pencils.
- Graphite is also used as a moderator in nuclear reactors, due to its ability to slow down neutrons.

### Fullerene

- Fullerenes are a class of carbon molecules in which the carbon atoms are arranged in a spherical or cylindrical shape.
- They have unique physical and chemical properties, and have potential applications in electronics, medicine, and materials science.
- The most well-known fullerene is the buckminsterfullerene, or “buckyball,” which is a spherical molecule composed of 60 carbon atoms.

## Principles of Green Chemistry

Green chemistry is the approach in chemical sciences that efficiently uses renewable raw materials, eliminates waste, and avoids the use of toxic and hazardous reagents and solvents in the manufacture and application of chemical products. There are twelve basic principles of Green Chemistry that can be grouped into “Reducing Risk” and “Minimizing the Environmental Footprint”.

1. **Prevention of waste or by-products:** According to this principle of green chemistry, priority is given to the prevention of waste rather than cleaning up and treating waste after it has been generated .
2. **Atom Economy:** This principle focuses on designing synthetic methods to maximize the incorporation of all materials used in the process into the final product .
3. **Reduce derivatives:** One of the key principles of green chemistry is to reduce the use of derivatives and protecting groups in the synthesis of target molecules. One of the best ways of doing this is the use of enzymes. Enzymes are so specific that they can often react with one site of the molecule and leave the rest of the molecule alone and hence protecting it .
4. **Design for energy efficiency:** Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure .
5. **Use of renewable feedstocks:** A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable .
6. **Design for degradation:** Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment .

### K3

1. K3 is a term that can refer to several different things, depending on the context in which it is used.
2. In mathematics, K3 can refer to a K3 surface, which is a complex algebraic surface with trivial canonical bundle.
3. In music, K3 is a Belgian-Dutch girl group that was formed in 1998.
4. In mountaineering, K3 is the name given to the mountain Skil Brum, which is located in the Karakoram range in Pakistan.
5. In transportation, K3 can refer to a class of steam locomotives used by the London and North Eastern Railway in the United Kingdom.
6. In biology, K3 can refer to a type of potassium channel that is found in the nervous system of mammals.
7. In education, K3 can refer to the third year of kindergarten, which is typically for children who are 4-5 years old.

### CO-2 Apply the fundamental concepts of determination of structure with various spectral techniques

1. Spectroscopy is the study of the interaction between matter and electromagnetic radiation.
2. Spectral techniques are used to determine the structure of molecules by analyzing the interaction of electromagnetic radiation with the molecule.

3. Some common spectral techniques include infrared (IR) spectroscopy, nuclear magnetic resonance (NMR) spectroscopy, and mass spectrometry (MS).
4. In IR spectroscopy, the molecule absorbs infrared radiation, causing vibrations in the chemical bonds. The resulting spectrum can be used to identify functional groups and determine the structure of the molecule.
5. In NMR spectroscopy, the molecule is placed in a magnetic field and exposed to radiofrequency radiation. The resulting spectrum can be used to determine the chemical environment of the nuclei in the molecule and provide information about the structure of the molecule.
6. In MS, the molecule is ionized and the resulting fragments are separated based on their mass-to-charge ratio. The resulting spectrum can be used to determine the molecular weight and structure of the molecule.
7. These techniques can be used in combination to provide a more complete picture of the structure of the molecule. For example, IR and NMR spectroscopy can be used to identify functional groups and determine the chemical environment of the nuclei, while MS can provide information about the molecular weight and fragmentation pattern of the molecule.

## Stereochemistry

Stereochemistry is a subdiscipline of chemistry that involves the study of the relative spatial arrangement of atoms that form the structure of molecules and their manipulation. It is also known as 3D chemistry, where the prefix “stereo” means “three-dimensionality”.

The study of stereochemistry focuses on the relationships between stereoisomers, which by definition have the same molecular formula and sequence of bonded atoms. This property is called chirality.

Some important concepts in stereochemistry include:

- Chirality: A molecule is chiral if it is not superimposable on its mirror image.
- Stereoisomers: Molecules with the same molecular formula and sequence of bonded atoms, but differ in the three-dimensional orientations of their atoms in space.
- Enantiomers: Stereoisomers that are non-superimposable mirror images of each other.
- Diastereomers: Stereoisomers that are not mirror images of each other.
- Assigning configurations using the R,S system: A system used to distinguish between enantiomers.
- Optical activity: The ability of a chiral molecule to rotate the plane of polarized light.
- Fischer projections: A two-dimensional representation of a three-dimensional molecule.

## K4

K4 is a term that can refer to several different things. Without more context, it is difficult to determine which specific K4 the user is referring to. Some possible interpretations of K4 include:

1. K4 is a type of visa for certain children of K-3 visa holders.
2. K4 is a type of cipher machine used by the German military during World War II.
3. K4 is a type of locomotive used by the Pennsylvania Railroad.
4. K4 is a type of protein kinase enzyme.
5. K4 is a type of mountain in the Karakoram range.

## CO-3 Utilize the theory of construction of electrodes, batteries and fuel cells in redesigning new engineering

1. **Electrodes:** Electrodes are conductive materials that allow the flow of electric current in and out of a cell or battery. They are typically made of metal or carbon and are designed to have a high surface area to facilitate the transfer of electrons.
2. **Batteries:** Batteries are devices that store electrical energy in the form of chemical energy. They consist of one or more electrochemical cells, each of which contains two electrodes separated by an electrolyte. The chemical reactions that occur within the cell generate a flow of electrons from one electrode to the other, producing an electric current.
3. **Fuel Cells:** Fuel cells are electrochemical devices that convert the chemical energy of a fuel directly into electricity. Like batteries, they consist of an anode, a cathode, and an electrolyte. However, unlike batteries, fuel cells require a continuous supply of fuel and oxygen to generate electricity.
4. **Redesigning:** The theory of construction of electrodes, batteries, and fuel cells can be utilized in redesigning new engineering solutions. This can involve improving the efficiency and performance of existing technologies, or developing entirely new systems that take advantage of the latest advances in materials science and electrochemistry.

## Corrosion of Products

Corrosion is the process by which a material deteriorates due to a chemical reaction with its environment. This can lead to the weakening of the material and ultimately its failure. There are several reasons for corrosion and methods to control it.

### Reasons for Corrosion

1. **Chemical reactions:** Corrosion is caused by chemical reactions between the material and its environment. For example, iron reacts with oxygen and water to form rust.



2. **Electrochemical reactions:** Corrosion can also occur due to electrochemical reactions, where electrons are transferred between the material and its environment. This can happen when two different metals are in contact with each other in the presence of an electrolyte.
3. **Environmental factors:** The environment can play a role in corrosion. For example, high humidity, high temperatures, and the presence of pollutants can accelerate the corrosion process.

## Methods to Control Corrosion

1. **Material selection:** One way to control corrosion is to select materials that are resistant to corrosion in the specific environment in which they will be used.
2. **Coatings:** Another way to control corrosion is to apply coatings to the material. These coatings can act as a barrier between the material and its environment, preventing or slowing down the corrosion process.
3. **Cathodic protection:** Cathodic protection is a method used to control corrosion in which a sacrificial anode is used to protect the material. The anode corrodes preferentially, protecting the material from corrosion.
4. **Environmental control:** Controlling the environment can also help to control corrosion. For example, reducing humidity, temperature, and the presence of pollutants can slow down the corrosion process.

In conclusion, corrosion is a process that can weaken and ultimately cause the failure of materials. There are several reasons for corrosion and methods to control it. By understanding the reasons for corrosion and implementing methods to control it, we can develop products that are more resistant to corrosion.

## Understanding of Chemistry of Engineering materials (Cement)

- Cement is an adhesive substance used in building and civil engineering construction.
- There are two kinds of cement based on the way it is set and hardened: hydraulic cement and non-hydraulic cement.
- Hydraulic cement hardens due to the addition of water, while non-hydraulic cement hardens by carbonation with the carbon present in the air.
- Portland cement, a type of hydraulic cement, is made up of four main compounds: tricalcium silicate, dicalcium silicate, tricalcium aluminate, and a tetra-calcium aluminoferrite.
- The chemical composition of cement will depend on the raw materials used in cement manufacturing, which are lime, silica, alumina, and iron oxide.
- Cement may be used alone or mixed with inert material known as aggregate to form mortar and concrete.

- With a better understanding of cement chemistry, scientists could potentially alter production or change ingredients to reduce emissions or add ingredients that are capable of actively absorbing carbon dioxide.

### K3

- K3 is a term that can refer to several different things, depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 can refer to a K3 surface, which is a complex algebraic surface with certain properties.
- In music, K3 is a Belgian-Dutch girl group.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

### CO-4 Develop understanding of the sources, impurities and hardness of water, apply the concepts of

1. **Sources of water:** Water can be obtained from various sources such as rivers, lakes, oceans, groundwater, and rainwater.
2. **Impurities in water:** Water from natural sources often contains impurities such as dissolved minerals, organic matter, and microorganisms. These impurities can affect the taste, odor, and appearance of the water, as well as its suitability for certain uses.
3. **Hardness of water:** Hardness refers to the concentration of calcium and magnesium ions in water. Hard water can cause problems such as scaling in pipes and reduced effectiveness of soaps and detergents.
4. **Concepts of water treatment:** Water treatment processes such as filtration, sedimentation, and disinfection can be used to remove impurities and reduce the hardness of water. These processes can improve the quality of water for drinking and other uses.

## Determination of Calorific Values and Analysis of Coal

Calorific value is a measure of the amount of energy produced from a unit weight of coal when it is combusted in oxygen. A measured sample of coal is completely combusted in a bomb calorimeter, which is a device for measuring heat. Standard bomb calorimeters can be used to determine the gross calorific value and are described in the ISO 1928 and ASTM D5865 standard. The two categories of calorimeters for coal analysis are adiabatic and isoperibolic .

Elemental or ultimate analysis encompasses the quantitative determination of carbon, hydrogen, nitrogen, sulfur, and oxygen within the coal. Additionally,

specific physical and mechanical properties of coal and particular carbonization properties are also determined. The calorific value of coal is the heat liberated by its complete combustion with oxygen.

Moisture content is an important parameter in coal analysis. It is needed for determining the calorific (heating) value and handling properties of a coal. The calorific value mainly comes from the combustion of organic components (C, H, N). In addition, moisture and ash in coal will absorb part of the heat during combustion .

### **K3**

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific meaning of K3 the user is referring to. Some possible meanings of K3 include:

1. K3 surface: In mathematics, a K3 surface is a complex algebraic surface with trivial canonical bundle.
2. K3 (mountain): K3 is the name of a mountain in the Karakoram range, also known as Broad Peak.
3. K3 (band): K3 is a Belgian-Dutch girl group with a Dutch repertoire, primarily enjoyed by pre-adolescent children.
4. K-3 (education): K-3 is a term used in education to refer to the first four years of primary education, typically covering kindergarten through third grade.
5. K3 (tax form): K3 is a tax form used in Sweden for reporting income from business activities.

### **CO-5 Develop the understanding of Chemical structure of polymers and its effect on their various properties**

1. Polymers are large molecules made up of repeating units called monomers.
2. The chemical structure of a polymer refers to the arrangement of its monomers and the type of bonds that hold them together.
3. The chemical structure of a polymer affects its physical and mechanical properties, such as its strength, elasticity, and melting point.
4. The type of bonds between monomers can affect the polymer's properties. For example, polymers with covalent bonds between monomers are generally stronger and more rigid than those with weaker bonds, such as hydrogen bonds or van der Waals forces.
5. The arrangement of monomers can also affect the polymer's properties. For example, linear polymers, where monomers are arranged in a straight chain, are generally stronger and more rigid than branched polymers, where monomers are arranged in a branched structure.
6. The presence of functional groups, such as carboxyl or amino groups, can also affect the polymer's properties by introducing additional bonding or interactions between monomers.

7. The degree of polymerization, or the number of monomers in a polymer chain, can also affect its properties. Generally, as the degree of polymerization increases, the polymer becomes stronger and more rigid.
8. In summary, the chemical structure of a polymer, including the type of bonds between monomers, the arrangement of monomers, the presence of functional groups, and the degree of polymerization, can all affect its physical and mechanical properties. Understanding these relationships is important for the design and development of new polymers with desired properties.

## Applications of Specific Polymers and Chemistry

Polymers are incredibly diverse elements that represent such fields of engineering from avionics through biomedical applications, drug delivery system, biosensor devices, tissue engineering, cosmetics, etc. The application of polymers and their subsequent composites is still advancing and increasing quickly due to their ease regarding manufacturing.

Some of the industries in which you would see the use and application of various polymeric materials and polymers themselves are: 1. In aircraft, aerospace, and sports equipment 2. 3D printing plastics 3. Biopolymers in molecular recognition 4. Polymers in holography 5. Organic polymers used in water purification 6. Printed circuit board substrates 7. Green Chemicals: Polymers and Biopolymers 8. Polymeric Biomolecules

Man-made polymers are used to make materials like rubber, plastics, glass, paper, and concrete. As per Polymer Chemistry, the size of polymers range from thousands to millions of atomic mass units, which causes them to have peculiar physical and chemical properties.

Polymer chemists have designed and synthesized polymers that vary in hardness, flexibility, softening temperature, solubility in water, and biodegradability. They have produced polymeric materials that are as strong as steel yet lighter and more resistant to corrosion.

Sample research topics that span the journal's scope are polymer applications in energy storage and conversion, separations, membranes, adhesives, functional coatings, sensing, adaptive and reconfigurable materials, electronics, photonics, biomaterials, and nanocomposites.

## Industrial Processes

Industrial processes are procedures involving chemical, physical, electrical, or mechanical steps to aid in the manufacturing of an item or items, usually carried out on a very large scale. These processes are critical to the production of goods used in everyday life and are applicable in a wide range of industries, including:

1. **Chemical Industry:** Industrial processes are used to convert raw materials into chemicals for use in various products, such as plastics, fertilizers, and pharmaceuticals.
2. **Food Industry:** Industrial processes are used to produce, process, and package food and beverages for consumption.
3. **Textile Industry:** Industrial processes are used to produce fibers, yarns, and fabrics for use in clothing and other textile products.
4. **Automotive Industry:** Industrial processes are used to manufacture vehicles and their components, such as engines, transmissions, and body parts.
5. **Electronics Industry:** Industrial processes are used to produce electronic components and devices, such as semiconductors, circuit boards, and consumer electronics.

These are just a few examples of the many industries that rely on industrial processes to produce the goods and services that we use every day. Industrial processes are constantly evolving and improving to increase efficiency, reduce waste, and minimize environmental impact.

### K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.

### Reference Books:

- Reference books are a type of non-fiction book that provides factual information on a particular subject.
- They are often used for research, study, or to gain a general understanding of a topic.
- Some common types of reference books include dictionaries, encyclopedias, almanacs, and atlases.
- Reference books are usually organized in a way that makes it easy to find the information you need, such as alphabetically or by subject.
- They are often updated regularly to ensure that the information they contain is accurate and up-to-date.
- Reference books can be found in libraries, bookstores, and online.
- Some popular reference book publishers include Oxford University Press, Cambridge University Press, and Merriam-Webster.

- It is important to evaluate the credibility of a reference book before using it as a source of information. This can be done by checking the publisher, the author's credentials, and the date of publication.

### **Engineering Chemistry by Rath & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi**

- Engineering Chemistry covers all topics taught to the first-year undergraduate students of all engineering disciplines offered by various Indian universities.
- The book presents each chapter in a simple, comprehensive, and illustrative manner.
- It covers topics from diverse branches of chemistry, viz., general, inorganic, organic, and analytical chemistry in addition to a topic, nanomaterials, from material science .
- Each chapter includes a variety of solved and unsolved problems/exercises to help reinforce learning and boost the confidence of the learner/reader.
- The specific focus in this book is on the syllabi of Biju Patnaik University of Technology (BPUT), KIIT (Deemed University), and SOA (Deemed University) in the state of Orissa.

### **2. Engineering Chemistry by SS Dara, S Chand & Co Ltd**

- Engineering Chemistry is a textbook written by SS Dara and published by S Chand & Co Ltd.
- The book is intended for students pursuing engineering courses and covers the fundamental concepts of chemistry as applied to engineering.
- The book is divided into several chapters, each covering a specific topic in the field of engineering chemistry.
- Some of the topics covered in the book include atomic structure, chemical bonding, states of matter, chemical thermodynamics, electrochemistry, and corrosion.
- The book also includes numerous examples, illustrations, and exercises to help students understand the concepts better.
- Engineering Chemistry by SS Dara is a popular choice among engineering students and is widely recommended by professors and instructors.

### **3. Engineering Chemistry by Jain & Jain, S.Chand & Comp, New Delhi**

- Engineering Chemistry is a textbook written by Jain & Jain and published by S.Chand & Comp, New Delhi.
- The book covers the fundamental concepts of chemistry and their applications in engineering.
- The topics covered in the book include atomic structure, chemical bonding, states of matter, chemical thermodynamics, electrochemistry, and more.

- The book is written in a clear and concise manner, making it easy for students to understand the concepts.
- The book also includes numerous examples and practice problems to help students apply the concepts they have learned.
- Engineering Chemistry by Jain & Jain is a popular choice among engineering students and is widely used as a reference book for various engineering courses.

#### **4. Engineering Chemistry by K. Sesha Maheswaramma, Pearson**

- “Engineering Chemistry” is a textbook written by K. Sesha Maheswaramma and published by Pearson.
- The book covers fundamental concepts in chemistry and their applications in engineering.
- Topics covered in the book include atomic structure, chemical bonding, phase rule, solutions, electrochemistry, corrosion, and water technology.
- The book is designed for undergraduate students in engineering and provides a comprehensive introduction to the subject.
- It includes numerous examples, illustrations, and exercises to aid in understanding and application of the concepts.
- The book is well-organized and written in a clear and concise manner, making it an excellent resource for students and professionals alike.

#### **5. Engineering Chemistry by OG Palanna, Mc Graw Hill Education, New Delhi**

- “Engineering Chemistry” is a book written by O.G. Palanna and published by Tata McGraw Hill Publishing Pvt.Ltd. in New Delhi in 2009 .
- The book is meant for first-year undergraduate students of engineering and provides a comprehensive exposure to the fundamentals of Engineering Chemistry .
- The presentation is lucid, the language is simple, and the flow of topics is logical, along with relevant concepts that help students acquire a good command of the basics of the subject .
- The book has received positive reviews for its content and presentation .

#### **6. Engineering Chemistry by Shashi Chawala, Dhanpat Rai Publishing Comp, New Delhi**

- Engineering Chemistry is a textbook written by Shashi Chawala and published by Dhanpat Rai Publishing Company, New Delhi.
- The book is intended for undergraduate students in engineering and covers the fundamental concepts of chemistry as applied to engineering.
- The book is divided into several chapters, each covering a specific topic in the field of engineering chemistry.
- Some of the topics covered in the book include atomic structure, chemical bonding, states of matter, chemical thermodynamics, electrochemistry,

and corrosion.

- The book also includes numerous examples, illustrations, and exercises to help students understand the concepts presented.
- Engineering Chemistry by Shashi Chawala is a widely used textbook in engineering programs and is considered a valuable resource for students studying engineering chemistry.

#### **7. University Chemistry by BH Mahan**

- University Chemistry by BH Mahan is a widely used textbook for undergraduate chemistry courses.
- The book covers a broad range of topics in general chemistry, including atomic structure, chemical bonding, thermodynamics, and kinetics.
- It is known for its clear explanations and well-organized presentation of material.
- The book includes numerous examples and practice problems to help students understand and apply the concepts.
- It is a valuable resource for students studying chemistry at the university level.

#### **8. University Chemistry by CNR Rao**

- University Chemistry is a book written by C.N.R. Rao and published by Macmillan Co. in 1973. The book has been reprinted several times since its first publication .
- The book provides an integrated approach into the principles and practice of chemistry and chemical sciences .
- C.N.R. Rao is a renowned scientist who focuses on Inorganic chemistry, Crystallography, Condensed matter physics, Nanotechnology and Metal. His research integrates issues of Oxide, Nanoparticle, Catalysis, Analytical chemistry and Amine gas treating in his study of Inorganic chemistry .
- Rao has also written other books on chemistry, including “Experiments in General Chemistry” with U.C. Agarwala, published by East-West Press, New Delhi (Van Nostrand-Reinhold) .
- Rao has established the International Centre for Materials Science (ICMS) which offers the C N R Rao Prize Lecture in Advanced Materials since 2010. The World Academy of Sciences instituted the TWAS-C.N.R. Rao Award for Scientific Research since 2006 for scientists in the least developed countries .